What Happens When We Enter a URL?

From Browser to Server

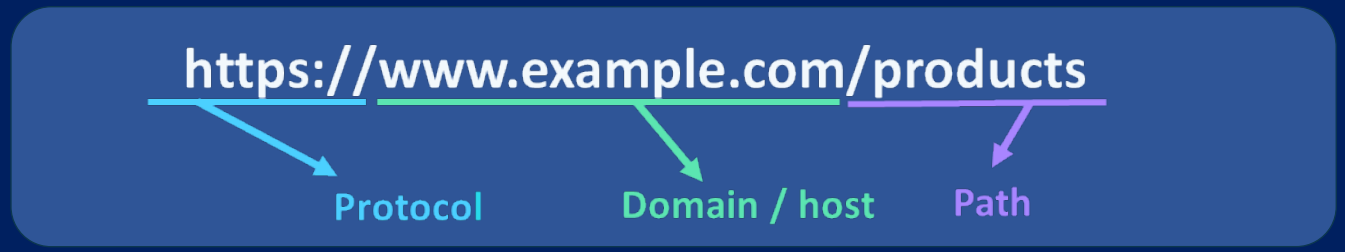


Technical English Project

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A simple address hides a few steps

When we press Enter, the browser finds the destination, requests the page, and displays the result.



DNS is like looking up a contact by name.

It helps computers find the right destination without us memorizing numbers.

URL (Uniform Resource Locator)

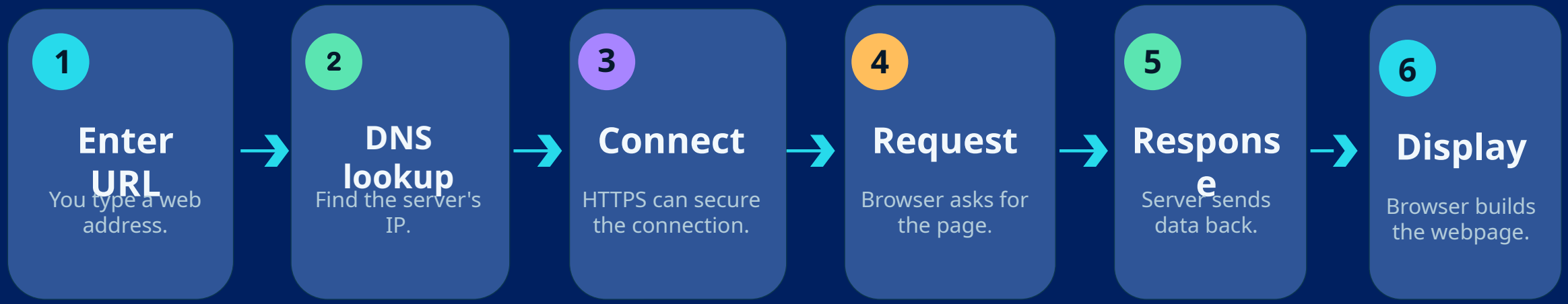
The address used to locate a resource on the web

5 technical terms

Browser	Opens websites.
DNS	Finds a domain's IP address.
IP address	A numeric network address.
HTTP / HTTPS	Rules for web communication.
Web server	Receives requests and sends data.

From Enter to a loaded page

The main steps — simplified

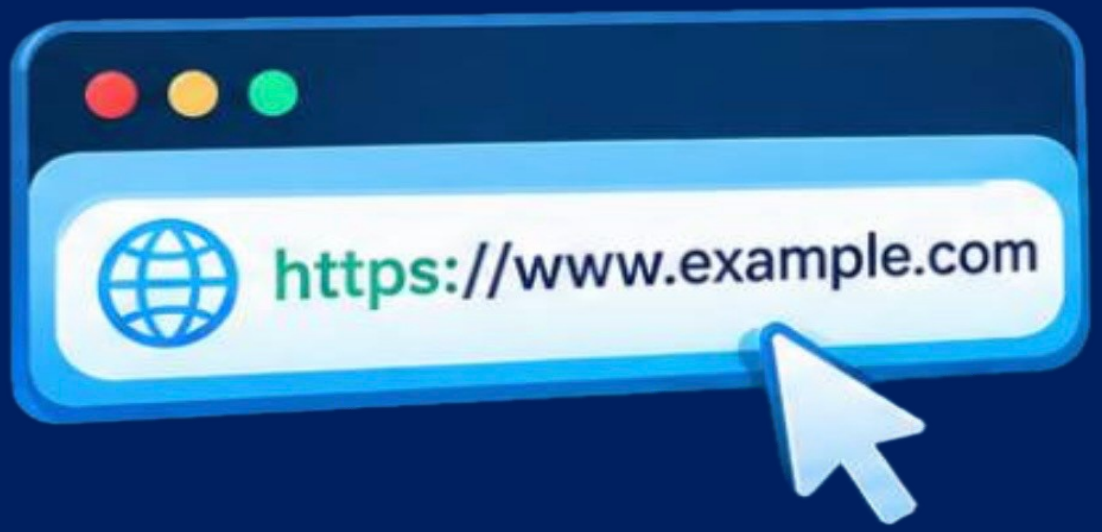


URL → DNS / IP address → HTTPS connection → HTTP request → response → webpage

Note: DNS information may already be saved in a cache, so a full lookup is not always needed.

1.Entering the URL

When you type a URL into a browser and press Enter,the browser analyzes the address to identify the destination and the resource we want to access.





The browser may check if it already has the website's information in its cache (temporary storage). example:

2. DNS Lookup

Computers use IP addresses to communicate, but DNS translates domain names like google.com into IP addresses.



Typical steps:

1. Check the browser/OS cache
2. Check the local hosts Ale
3. Query a DNS server (recursive → authoritative)

3. Secure Connection (HTTPS)

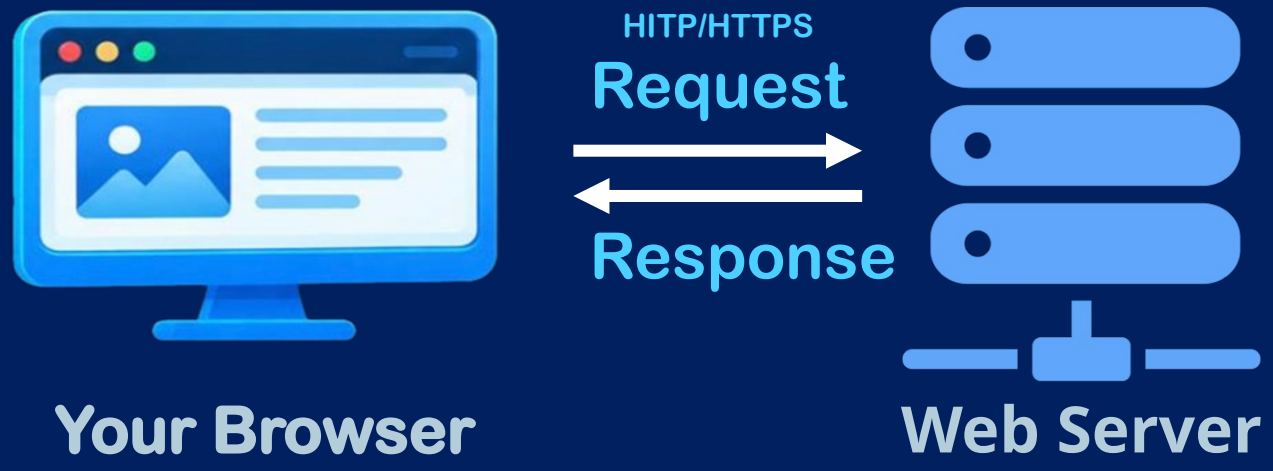
After getting the IP address, your browser establishes a secure connection with the server using HITPS (SSL/TLS)



✅ Authenticates the server
✅ Encrypts the data
✅ Protects your privacy)

4/5. Request & Response

The browser sends an HTTP request to the server. The server sends back an HTTP response, which may include HTML, CSS, JavaScript, images, and other resources.



GET /products/ HTTP/1.1
host:www.example.com
User-Agent: Chrome/9...

6. Website Appears

The browser processes the website's code and displays the web page.



A product page is requested

Example only — a real exchange may include additional headers and data.

BROWSER → SERVER

GET /products

```
Host: www.example.com
```

The browser asks the server for the products page.



SERVER → BROWSER

200 OK

Content: HTML + other resources

The server responds; the browser uses the content to build the page.

200 = success • **404 = resource not found** • **500 = Internal Server Error**

Three things to remember

01	**URL**	It tells the browser which resource we want.
02	**DNS**	It helps find the IP address for a domain.
03	**HTTP / HTTPS**	The browser and server exchange requests and responses.

Quick questions for the class

1. Why do we need DNS instead of using domain names alone?

Hint: think about how computers find each other.

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2. What is the difference between a browser and a web server?

Hint: one requests and displays; the other responds.

In short: the browser finds the destination, asks for data, and turns the response into a webpage.

Sources & further reading

These are English-language technical references for checking the main ideas.

1. **MDN Web Docs — How the web works**
   https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Environment_setup/How_the_web_works

2. **MDN Web Docs — What is a URL?**
   https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Web_mechanics/What_is_a_URL

3. **Cloudflare Learning Center — What is DNS?**
   https://www.cloudflare.com/learning/dns/what-is-dns/

Thank you for listening!